How-To Library

Informatica

Enabling SAML Authentication with PingFederate in Informatica 10.4.0

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Abstract

You can enable users to log into Informatica web applications using single sign-on. This article explains how to configure single sign-on in an Informatica 10.4.0 domain using Security Assertion Markup Language (SAML) and the PingFederate identity provider.

Supported Versions

- Informatica Data Engineering Integration 10.4.0
- Informatica Enterprise Data Catalog 10.4.0
- Informatica Enterprise Data Preparation 10.4.0
- Informatica Metadata Manager 10.4.0

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Overview

You can configure Security Assertion Markup Language (SAML) authentication in an Informatica version 10.4.0 domain using the PingFederate identity provider.

Security Assertion Markup Language is an XML-based data format for exchanging authentication information between a service provider and an identity provider. In an Informatica domain, an Informatica web application is the service provider.

Informatica supports PingFederate version 9.2.2. You can configure the following Informatica web applications to use SAML authentication:

- Informatica Administrator
- Informatica Analyst
- Metadata Manager
- Enterprise Data Catalog
- Enterprise Data Preparation

Note: SAML authentication cannot be used in an Informatica domain configured to use Kerberos authentication.

SAML Authentication Process

Informatica web applications and the identity provider exchange authentication information to enable SAML authentication in an Informatica domain.

The following steps describe the basic SAML authentication flow:

1. A user accesses an Informatica web application.
2. The user selects the security domain containing LDAP user accounts used for SAML authentication on the application log in page, and then clicks the log in button.

   If the user selects the native security domain, the user provides a user name and password and logs in to the application.
3. Based on the identity provider configuration, the user is prompted to provide the credentials required for first time authentication.
4. The identity provider validates the user's credentials and creates a session for the user.

   The identity provider also validates the target web application URL, and then redirects the user to the web application with a SAML token containing the user's identity information.
5. The application validates the SAML token and user identity information, creates a user session, and then completes the user log in process.

The existing user session in the browser is used for subsequent authentication. To access another Informatica web application configured to use SAML authentication, the user selects the LDAP security domain on the application log in page. The user does not need to supply a user name or password.

The user remains logged in to all Informatica web applications that are running in the same browser session. However, if the user logs out of an Informatica web application, the user is also logged out of other Informatica web applications running in the same browser session.

Enable SAML Authentication in a Domain

Configure the identity provider, the Informatica domain, and the gateway nodes within the domain to use SAML authentication.

To configure SAML authentication for supported Informatica web applications that run in a domain, perform the following tasks:

1. Create an LDAP configuration to connect to the LDAP identity store that contains Informatica web application user accounts. You also create an LDAP security domain, and then import the user accounts into the security domain.
2. Export the assertion signing certificate from the identity provider.
3. Import the assertion signing certificate into a truststore file on each gateway node in the domain. You can import the certificate into the Informatica default truststore file, or into a custom truststore file.
4. Add one or more relying party trusts or service providers in the identity provider.
5. Add the URL for each Informatica web application to the identity provider.
6. Enable SAML authentication in the domain.
7. Enable SAML authentication on every gateway node in the domain.

Create a Security Domain

When you configure an Informatica domain to use PingFederate, you use Microsoft Active Directory as the identity store. Use the Administrator tool to create an LDAP configuration and a security domain for web application user

accounts stored in the identity store that contains user accounts, and then import each account from the identity store into the security domain.

You must import the LDAP accounts for all users that use SAML authentication into the security domain. After importing the accounts into the domain, assign the appropriate Informatica domain roles, privileges and permissions to the accounts within the LDAP security domain.

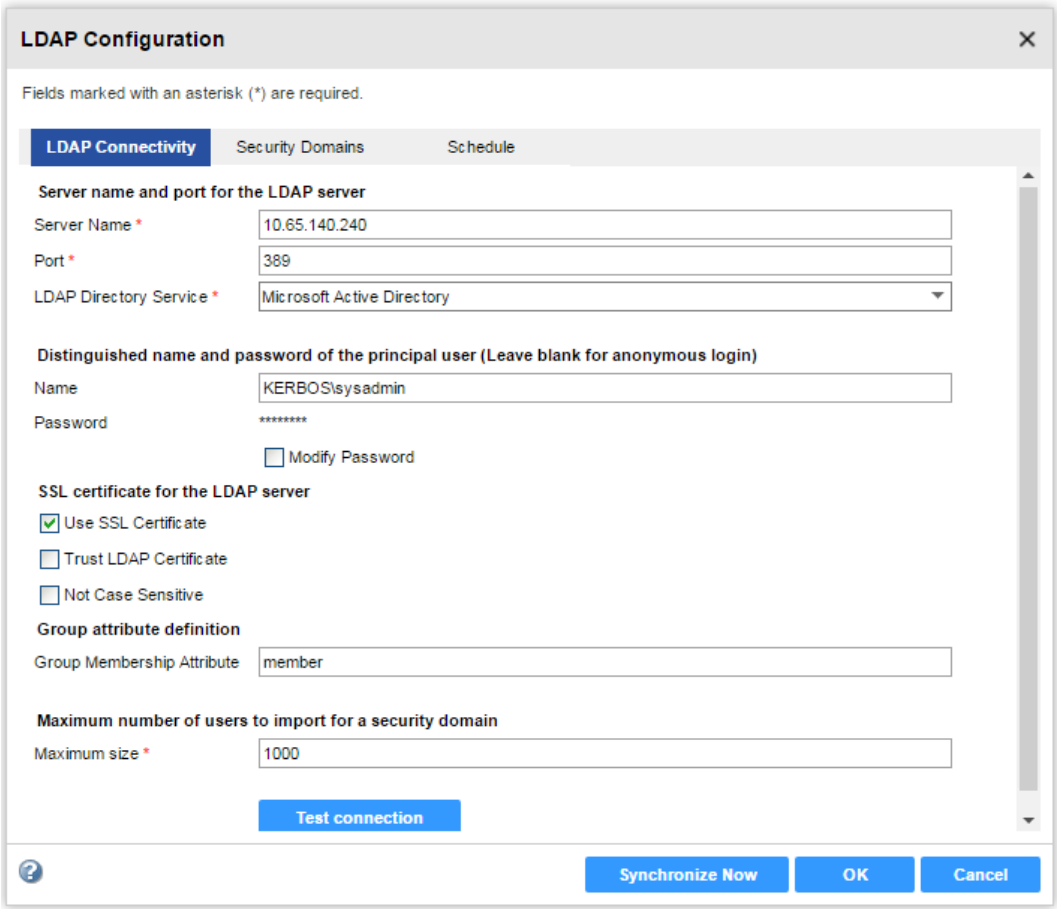
1. In the Administrator tool, click the **Users** tab, and then select the **Security** view.
2. Click the **Actions** menu and select **LDAP Configuration**.

   The **LDAP Configuration** dialog box opens.
3. Click the **LDAP Connectivity** tab.
4. Configure the connection properties for the Active Directory server.

   The following table describes the server connection properties:

Property	Description
Server Name	Host name or IP address of the Active Directory server.
Port	Listening port for the Active Directory server. The default value is 389.
LDAP Directory Service	Select **Microsoft Active Directory**.
Name	Distinguished name (DN) for the principal LDAP user. The user name often consists of a common name (CN), an organization (O), and a country (C). The principal user name is an administrative user with access to the directory. Specify a user that has permission to read other user entries in the directory service.
Password	Password for the principal LDAP user.
Use SSL Certificate	Indicates that the LDAP server uses the Secure Socket Layer (SSL) protocol. If the LDAP server uses SSL, you must import the certificate into a truststore file on every gateway node within the Informatica domain. You must also set the INFA_TRUSTSTORE and INFA_TRUSTSTORE_PASSWORD environment variables if you do not import the certificate into the default Informatica truststore.
Trust LDAP Certificate	Determines whether the Service Manager can trust the SSL certificate of the LDAP server. If selected, the Service Manager connects to the LDAP server without verifying the SSL certificate. If not selected, the Service Manager verifies that the SSL certificate is signed by a certificate authority before connecting to the LDAP server.
Not Case Sensitive	Indicates that the Service Manager must ignore case sensitivity for distinguished name attributes when assigning users to groups. Enable this option.
Group Membership Attribute	Name of the attribute that contains group membership information for a user. This is the attribute in the LDAP group object that contains the distinguished names (DNs) of the users or groups who are members of a group. For example, *member* or *memberof*.
Maximum size	Maximum number of user accounts to import into a security domain. If the number of user to be imported exceeds the value for this property, the Service Manager generates an error message and does not import any user. Set this property to a higher value if you have many users to import. The default value is 1000.

The following image shows the connection details for an LDAP server set in the LDAP Connectivity panel of the **LDAP Configuration** dialog box.

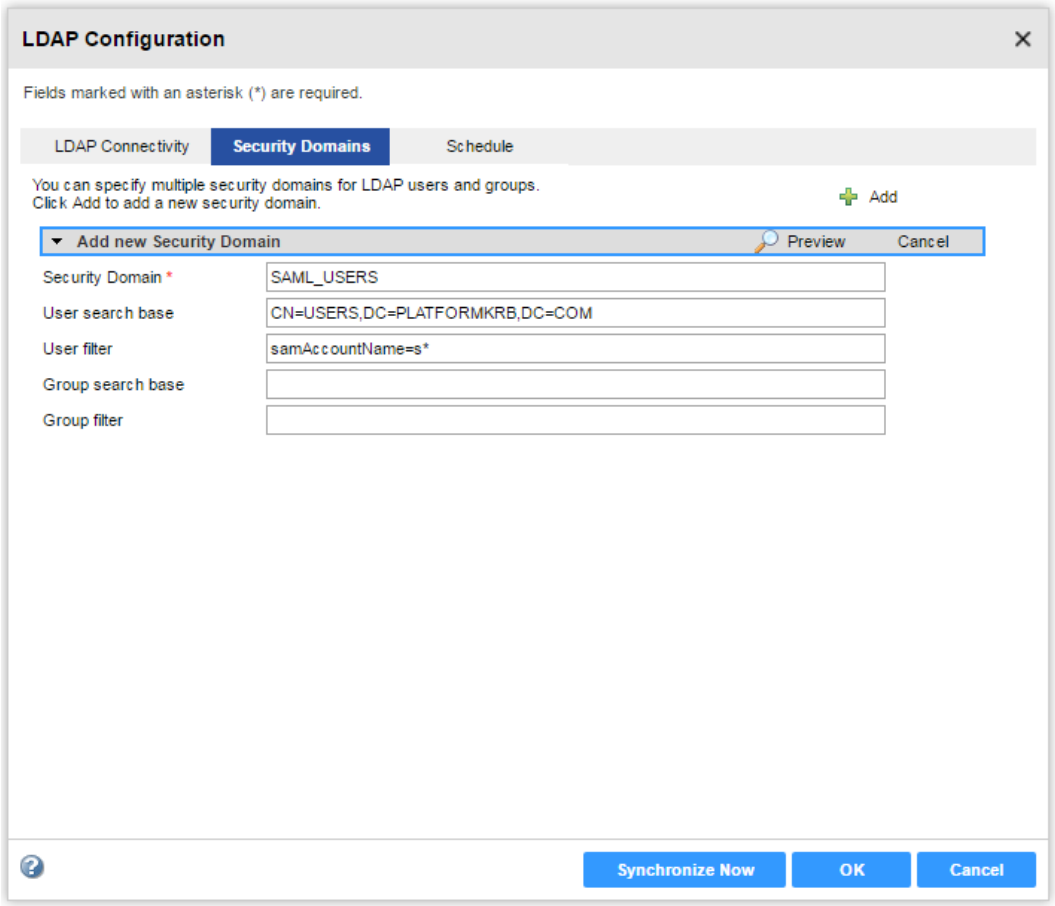


5. Click **Test Connection** to verify that the connection to the Active Directory server is valid.
6. Click the **Security Domains** tab.
7. Click **Add** to create a security domain.
8. Enter the security domain properties.

The following table describes the security domain properties:

Property	Description
Security Domain	Name of the LDAP security domain. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters or contain the following special characters: , + / < > @ ; \ % ? The name can contain an ASCII space character except for the first and last character. All other space characters are not allowed.
User search base	Distinguished name (DN) of the entry that serves as the starting point to search for user names in the LDAP directory service. The search finds an object in the directory according to the path in the distinguished name of the object. In Active Directory, the distinguished name of a user object might be cn=UserName,ou=OrganizationalUnit,dc=DomainName, where the series of relative distinguished names denoted by dc=DomainName identifies the DNS domain of the object. For example, to search the Users container that contains user accounts in the example.com Windows domain, specify CN=USERS,DC=EXAMPLE,DC=COM.
User filter	An LDAP query string that specifies the criteria for searching for users in Active Directory. The filter can specify attribute types, assertion values, and matching criteria. For Active Directory, format the query sting as: `sAMAccountName=<account>`
Group search base	Distinguished name (DN) of the entry that serves as the starting point to search for group names in Active Directory.
Group filter	An LDAP query string that specifies the criteria for searching for groups in the directory service.

The following image shows the properties for an LDAP security domain named SAML_USERS set in the Security Domains panel of the **LDAP Configuration** dialog box. The user filter is set to import all users beginning with the letter "s".



9. Click **Synchonize Now**.

   The security domain appears in the Users view.
10. Expand the domain in the Navigator to view the imported user accounts.
11. Set the appropriate roles, privileges, and permissions on the user accounts that will access each web application.

Import the Certificate into the Truststore Used for SAML Authentication

Import the assertion signing certificate used by the identity provider into the truststore file used for SAML authentication on every gateway node within the Informatica domain.

You can import the certificate into the default Informatica truststore file, or into a custom truststore file.

The file name of the default Informatica truststore file is `infa_truststore.jks`. The file is installed in the following location on each node:

```
<Informatica installation directory>\services\shared\security\infa_truststore.jks
```

Note: Do not replace the default `infa_truststore.jks` file with a custom truststore file.

If you import the certificate into a custom truststore file, you must save the truststore file in a different directory than the directory containing the default Informatica truststore file. The truststore file name must be `infa_truststore.jks`.

You can use the Java keytool key and certificate management utility to create an SSL certificate or a certificate signing request (CSR) as well as keystores and truststores in JKS format. The keytool is available in the following directory on domain nodes:

```
<Informatica installation directory>\java\bin
```

If the domain nodes run on AIX, you can use the keytool provided with the IBM JDK to create an SSL certificate or a Certificate Signing Request (CSR) as well as keystores and truststores.

1. Copy the certificate files to a local folder on a gateway node within the Informatica domain.
2. From the command line, go to the location of the keytool utility on the node.
3. Run the keytool utility to import the certificate.
4. Restart the node.

Create a Service Provider in PingFederate

Use the PingFederate administrative console to create a new Service Provider in PingFederate.

1. Click the **Identity Provider** tab, and then click the **Create New** button under the SP Connections header.
2. Specify the service provider identifier when you enable SAML in the domain.

   If you specify "Informatica" as the SP Connection name, you do not need to provide a value.
3. Click the **Create New** button.
4. Click the **Connection Type** tab, and then select **Protocol SAML 2.0** as the Browser SSO Profiles.
5. Click the **Connection Options** tab, and then select **Browser SSO**.
6. Click the **General Info** tab, then enter the required values.
7. Click **Next**.

Configure Browser SSO

Use the PingFederate administrative console to configure secure browser-based SSO.

1. On the Configure Browser SSO page, click the **SAML Profiles** tab, and then select **IDP-Initiated SSO** and **SP-Initiated SSO** under Single Sign On (SSO) Profiles.
2. Click the **Assertion Lifetime** tab, and then select the default value.
3. Click the **Assertion Creation** tab, and then click the **Configure Assertion Creation** button.
4. Click the **Identity Mapping** tab, and then select **STANDARD**.
5. Click the **Attribute Contract** tab, and then specify the following values:

Property	Value
SAML_SUBJECT	urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified
Extend Contract	SAML_AUTH_CTX - urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified
Username	urn:oasis:names:tc:SAML:1.1:nameid-format:unspecified

6. Click the **Authentication Source Mapping** tab.

7. Click the **Map New Adapter Instance** button.
8. Click the **Adapter Instance** tab.
9. Select **PingOne HTML** as the Adapter Instance Name.
10. Click the **Map New Adapter Instance** button.
11. Click the **Manage Adapter Instances** button.
12. Select **PingOne HTML** from the ADAPTER INSTANCE menu.
13. Select **policy.action** and **username** for ADAPTER CONTRACT.
14. Click the **Mapping Method** tab.
15. Select **USE ONLY THE ADAPTER CONTRACT VALUES IN THE SAML ASSERTION**.
16. Click the **Attribute Contract Fulfillment** tab.
17. Configure the following properties:

Attribute Contract	Source	Value
SAML_AUTH_CTX	Context	Authentication Context
SAML_SUBJECT	Adapter	username
username	Adapter	username

18. You do not need to configure properties on the Issuance Criteria page.
19. Click **Next**.

Configure the Protocol Settings

Use the PingFederate administrative console to add the endpoint URL for each Informatica web application you want to enable to use SAML authentication to PingFederate

1. On the Protocol Settings page, click the **Assertion Consumer Service URL** tab.
2. Enter the endpoint URL for an Informatica web application you want to enable to use SAML authentication.

   For the Administrator tool URL, append `Login.do` to the URL, as shown in the following example:

   ```
   https://server05.company.com:13345/administrator/Login.do
   ```

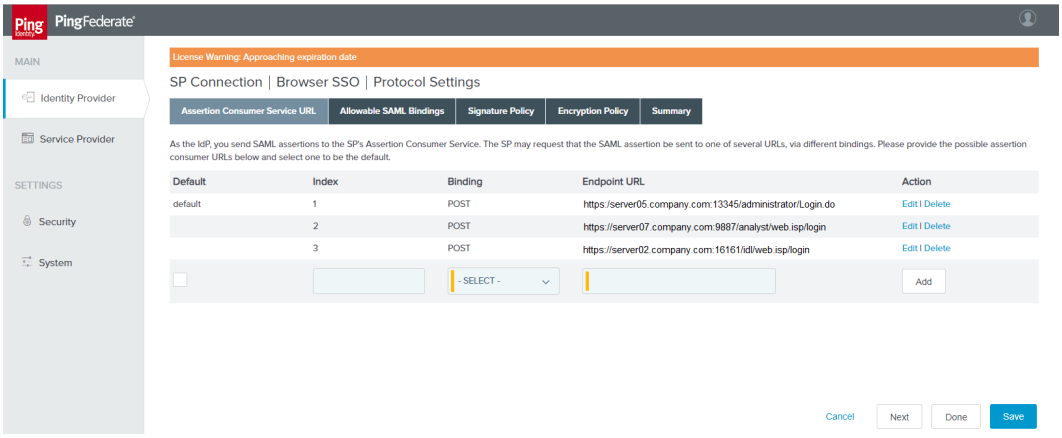
   For all other web applications, select the corresponding application service in the Administrator tool, and then copy the URL that is the value of the Service URL property. Append `web.isp/login` to the URL.

   The following example shows the URLs for the Analyst tool and for the Enterprise Data Preparation application:

   ```
   https://server07.company.com:9887/analyst/web.isp/login
   ```

   ```
   https://server02.company.com:16161/idl/web.isp/login
   ```

The following images shows the web application URLs added to the identity provider:



3. Select **POST** as the binding.
4. Click **Add**.

   Repeat this process for each web application.
5. Click the **Allowable SAML Bindings** tab.
6. Select the **POST** and **REDIRECT** options.
7. Click the **Signature Policy** tab.
8. Select the **ALWAYS SIGN ASSERTIONS** and **SIGN RESPONSE AS REQUIRED** options.
9. Click the **Encryption Policy** tab.
10. Select **NONE**.
11. Click **Save**.

Configure Credentials

Use the PingFederate administrative console to configure security credentials and complete the identity provider creation.

1. On the Configure Credentials tab, click the **Configure Credentials** button.
2. Click the **Digital Signature Settings** tab.
3. Configure the following properties:

Property	Value
Selected Certificate	Select the assertion signing certificate used by the identity provider.
Include Certificate in KeyInfo	true
Include Raw Key in KeyValue	true
Selected Signing Algorithm	RSA SHA256

4. Click **Next**.
5. Copy the Identity Provider URL. You specify the URL when you use infasetup commands to enable SAML authentication in the domain.

6. Click **Save**.

Enable SAML Authentication in the Domain

You can enable SAML authentication in an existing Informatica domain, or you can enable it when you create a domain.

When you enable a domain to use SAML authentication, all web applications that run in the domain use the default identity provider you specify when you enable SAML authentication in the domain. For example, if you configure AD FS as the identity provider, all web applications use AD FS as the identity provider, unless you configure a web application to use a different identity provider.

Select one of the following options:

Enable SAML authentication when you run Informatica installer.

You can enable SAML authentication and specify the identity provider URL when you configure the domain as part of the installation process.

Enable SAML authentication in an existing domain.

Use the infasetup updateDomainSamlConfig command to enable SAML authentication in an existing Informatica domain. You can run the command on any gateway node within the domain.

Enable SAML authentication when you create a domain.

Use the infasetup defineDomain command to enable SAML authentication when you create a domain.

See the *Informatica Command Reference* for instructions on using the commands.

infasetup DefineDomain Command Options

Use the infasetup defineDomain command to enable SAML authentication when you create a domain.

The following example shows the options to configure a domain to use PingFederate as the identity provider in the final six options at the command prompt:

```
infasetup defineDomain -cs "jdbc:company:oracle://host:1521;sid=DB2" -dt oracle -dn TestDomain -ad test_admin -pd test_admin -ld $HOME/ISP/1040/source/logs -nn TestNode1 -na host1.company.com -saml true -iu https://01.00.000.00:9031/idp/SSO.saml2 -spid PingFed_ProdDomain -cst 240 -asca pingcert -std \custom\security\ -stp password -mi 10000 -ma 10200 -rf $HOME/ISP/BIN/nodeoptions.xml
```

The following table describes the SAML options and arguments:

Option	Argument	Description
-EnableSaml -saml	true|false	Required. Set this value to true to enable SAML authentication for supported Informatica web applications within the Informatica domain. Set this value to false to disable SAML authentication for supported Informatica web applications within the Informatica domain.
-idpUrl -iu	identity_provider_url	Required. Specify the PingFederate identity provider URL for the domain. You must specify the complete URL string.
-ServiceProviderId -spid	service_provider_id	Optional. The relying service provider identifier for the domain as defined in PingFederate. If you specified "Informatica" as the service provider name, you do not need to specify a value.

Option	Argument	Description
-ClockSkewTolerance -cst	clock_skew_tolerance_in_seconds	Optional. The allowed time difference between the PingFederate host system clock and the master gateway node's system clock. The lifetime of SAML tokens issued by the identity provider by is set according to the host system clock. The lifetime of a SAML token issued by the identity provider is valid if the start time or end time set in the token is within the specified number seconds of the master gateway node's system clock. Values must be from 0 to 600 seconds. Default is 120 seconds.
-AssertionSigningCertificateAlias -asca	idp_assertion_signing_certificate_aliaseAlias	Required. The alias name specified when importing the identity provider assertion signing certificate into the truststore file used for SAML authentication.
-SamlTrustStoreDir -std	saml_truststore_directory	Optional. The directory containing the custom truststore file required to use SAML authentication on gateway nodes within the domain. Specify the directory only, not the full path to the file. SAML authentication uses the default Informatica truststore if no truststore is specified.
-SamlTrustStorePassword -stp	saml_truststore_password	Required if you use a custom truststore. The password for the custom truststore file.

See the *Informatica Command Reference* for instructions on using the infasetup defineDomain command.

Infasetup updateDomainSamlConfig Command Options

Set the SAML options in the infasetup updateDomainSamlConfig command to enable SAML authentication in a domain. Shut down the domain before you run the command.

Specify the identity provider URL as the value for the `-iu` option. The following example shows the command usage to configure a domain to use PingFederate as the identity provider:

```
infasetup updateDomainSamlConfig -saml true -iu https://01.00.000.00:9031/idp/SSO.saml2 -spid
PingFed_ProdDomain -cst 240
```

The following table describes the options and arguments:

Option	Argument	Description
-EnableSaml -saml	true|false	Required. Set this value to true to enable SAML authentication for supported Informatica web applications within the Informatica domain. Set this value to false to disable SAML authentication for supported Informatica web applications within the Informatica domain.
-idpUrl -iu	identity_provider_url	Required. Specify the PingFederate identity provider URL for the domain. You must specify the complete URL string.

Option	Argument	Description
-ServiceProviderId -spid	service_provider_id	Optional. The relying service provider identifier for the domain as defined in PingFederate. If you specified "Informatica" as the service provider name, you do not need to specify a value.
-ClockSkewTolerance -cst	clock_skew_tolerance_in_seconds	Optional. The allowed time difference between the PingFederate host system clock and the master gateway node's system clock. The lifetime of SAML tokens issued by the identity provider by is set according to the host system clock. The lifetime of a SAML token issued by the identity provider is valid if the start time or end time set in the token is within the specified number seconds of the master gateway node's system clock. Values must be from 0 to 600 seconds. Default is 120 seconds.

See the *Informatica Command Reference* for instructions on using the infasetup updateDomainSamlConfig command.

Enable SAML Authentication on the Gateway Nodes

You must configure SAML authentication on every gateway node in the Informatica domain.

Select one of the following options to configure SAML authentication on a gateway node:

Enable SAML authentication when you define a gateway node on a machine.

Use the infasetup DefineGatewayNode command to enable SAML authentication on the gateway node.

Enable SAML authentication when you configure a gateway node to join a domain that uses SAML authentication.

Use the infasetup UpdateGatewayNode command to enable SAML authentication on the gateway node.

Enable SAML authentication when you convert a worker node to a gateway node.

Use the isp SwitchToGatewayNode command to enable SAML authentication on the node.

See the *Informatica Command Reference* for instructions on using the commands.

Gateway Node Command Options

Use the infasetup DefineGatewayNode command to enable SAML authentication when you create a gateway node. Use infasetup UpdateGatewayNode or infacmd isp SwitchToGatewayNode to enable SAML authentication on an existing node.

The SAML options are identical for all of these commands. The following example shows the SAML options:

```
infasetup defineGatewayNode -cs "jdbc:informatica:oracle://host:1521;sid=xxxx" -du test_user -dp
test_user -dt oracle -dn TestDomain -nn TestNode1 -na host2.company.com:1234 -ld $HOME/ISP/1011/
source/logs -rf $HOME/ISP/BIN/nodeoptions.xml -mi 10000 -ma 10200 -ad test_admin -pd test_admin -
saml true -asca pingcert -std \custom\security\ -stp password
```

The following table describes the options and arguments:

Option	Description
-EnableSaml -saml	Required. Enables SAML authentication in the Informatica domain. Set this value to true to enable SAML authentication in the domain. Set this value to false to disable SAML authentication in the domain.
-AssertionSigningCertificateAlias -asca	Required if SAML authentication is enabled for the domain. The alias name specified when importing the identity provider assertion signing certificate into the truststore file used for SAML authentication.
-SamlTrustStoreDir -std	Optional. The directory containing the custom truststore file required to use SAML authentication on gateway nodes within the domain. Specify the directory only, not the full path to the file. The default Informatica truststore is used if no truststore is specified.
-SamlTrustStorePassword -stp	Required if you use a custom truststore. The password for the custom truststore file.

See the *Informatica Command Reference* for instructions on using the infasetup DefineGatewayNode, the infasetup UpdateGatewayNode, and the infacmd isp SwitchToGatewayNode commands.

Author

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